Daniel Cruz Gutiérrez

ENVIRONMENTAL DATA SCIENTIST — CLIMATE, ENERGY & GEOSPATIAL RISK

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RESEARCH PROFILE

My academic path has moved progressively from ecology to energy systems to climate and geospatial machine learning. I use statistical downscaling, neural networks, spatial statistics, and decision-making under uncertainty to study climate, energy, and infrastructure systems, and I bring hands-on experience with real satellite, climate-model, and land-parcel datasets rather than only classroom problems. I am seeking a fully funded PhD that builds on this trajectory — Earth observation, climate/infrastructure risk, or environmental data science — while deepening my mathematical and statistical foundations.

Research interests: climate & infrastructure risk · remote sensing & Earth observation · energy and infrastructure systems · statistical learning & decision-making under uncertainty · spatiotemporal modelling

EDUCATION

M.Sc. Energy and Sustainability 2024 – 2026

Pontificia Universidad Javeriana, Bogotá, Colombia — Faculty of Engineering

- Thesis: *"On the Edge of Uncertainty: Estimating the Impacts of Climate Change on the Hydroelectric Generation of the 'El Guavio' Dam in Cundinamarca, Colombia"* (advisor: Prof. Nelson Obregón Neira, in press). Statistically downscaled CMIP6 scenarios to the basin scale, trained a multilayer-perceptron neural network for reservoir inflow prediction, and used multidimensional clustering to derive operating rules.

B.Sc. Ecology 2020 – 2024

Pontificia Universidad Javeriana, Bogotá, Colombia — Faculty of Environmental and Rural Studies

- Focus on spatial analysis, environmental modelling, and quantitative ecology; foundation for later geospatial and statistical work.

RESEARCH & PROFESSIONAL EXPERIENCE

Data Scientist — Energy Research Jun. 2026 – Present

Aterio (Vancouver, Canada) — Remote, Bogotá

- Own the land-parcel and satellite change-detection data products over a proprietary inventory of 8,870 US data-center buildings, from acquisition through production pipelines and analyst review tooling — direct, applied experience in Earth observation and infrastructure monitoring at scale.
- Built a satellite change-detection review platform over Sentinel-2, Planet, Vantor, and Airbus imagery (Supabase edge functions, React/TypeScript UI), and diagnosed a defect where unbounded Sentinel-2 queries caused every run to compare an image against itself.
- Designed a Python parcel-acquisition pipeline with three-layer deduplication (coordinate matching, purchase history, R-tree point-in-polygon) against a per-record-billed API, cutting billed lookups 43–93% below one call per site.
- Resolved corporate ownership across single-purpose LLCs by normalizing mailing addresses, owner names, and tax-agent records, exposing 83 owner clusters at no additional API cost.
- Delivered a corpus of 1,170 US land parcels (121 attributes, 45 states) linked to 4,973 data-center buildings.

Research Intern
Dec. 2025 – Apr. 2026

Okinawa Institute of Science and Technology (OIST), Okinawa, Japan

- Configured and executed General Circulation Model (CESM) experiments on HPC clusters in Linux/Ubuntu environments, producing sea-surface-temperature, vertical-wind-shear, and solar-activity scenario outputs.
- Analyzed extreme-event behaviour — hurricane genesis, track, and intensity — across multi-model ensembles, benchmarked against observational datasets.

Research Associate
Jun. 2024 – Nov. 2025

Pontificia Universidad Javeriana, Bogotá, Colombia

- **Hydropower & climate risk** (M.Sc. thesis, ECOPAN / Civil Engineering): built and compared Random Forest, MLP neural network, fuzzy-logic, and clustering models for reservoir-inflow and dam-operation scenario analysis under CMIP6-downscaled climate projections.
- **Geospatial & remote sensing** (with IGAC and the Alexander von Humboldt Institute): automated Sentinel/Landsat workflows in Google Earth Engine — NDVI time series, Random Forest land-cover classification on CORINE Land Cover, radiometric correction — producing the human-footprint and landscape-connectivity analyses behind two published research chapters.
- **Climate extremes & public health** (Dept. of Civil Engineering): curated climate extreme indices (heatwave duration, extreme precipitation) in Python/R and correlated them with mortality data for respiratory and circulatory disease in Spain (manuscript in preparation).

Data Analyst Intern
Jan. – Aug. 2023

Brigard Urrutia, Bogotá, Colombia

- Built and quality-controlled the environmental and operational GIS datasets behind client risk assessments at a top-tier Colombian law firm, including a precipitation model for agricultural planning and a runoff-based flood-risk assessment.

INDEPENDENT RESEARCH PROJECTS

Self-directed portfolio built alongside professional work to prototype PhD research directions — full code at github.com/00danielcruz11-lang/phd-data-science-portfolio.

Water Catchment Decision Agent Under Uncertainty

- Prototype decision-making agent for reservoir operation under stochastic drought/flood hazard scenarios: a water-balance impact model coupled with a Markov decision process, solved by value iteration and evaluated by Monte Carlo policy simulation. Connects two research directions — generative modelling of hazard scenarios and optimization under uncertainty for climate adaptation.

Geospatial Human Footprint Analysis

- Composite human-pressure index built from VIIRS nighttime lights, distance-to-settlement and distance-to-road surfaces, land-cover reclassification, and neighborhood-texture fragmentation, implemented in Google Earth Engine and Python (geemap, rasterio).

PUBLICATIONS

1. Cruz Gutiérrez, D. & Correa Ayram, C. A. (2024). *Changes in the Configuration of the Human Spatial Footprint Over Time: A Prospective Analysis within the Framework of the Peace Agreement in the Department of Guaviare.* In *Geografías para la Vida*, Chapter 6. Instituto Geográfico Agustín Codazzi (IGAC).

2. Cruz, D., Correa Ayram, C., & Etter, A. (2024). *The Spatial Footprint in the Landscape of the Orinoquía.* Bio Report. Alexander von Humboldt Biological Resources Research Institute.

3. Cruz Gutiérrez, D. & Obregón Neira, N. *On the Edge of Uncertainty: Estimating the Impacts of Climate Change on the Hydroelectric Generation of the "El Guavio" Dam in Cundinamarca, Colombia.* M.Sc. Thesis, Pontificia Universidad Javeriana. *In press.*

Manuscript in preparation: extreme heat and public-health outcomes (respiratory and circulatory mortality) in Spain, with the Dept. of Civil Engineering, Pontificia Universidad Javeriana.

AWARDS & SCHOLARSHIPS

- Merit scholarship, M.Sc. in Energy and Sustainability, Pontificia Universidad Javeriana (2024).

TECHNICAL SKILLS

Programming: Python (pandas, GeoPandas, Shapely, scikit-learn, PyTorch, TensorFlow), R, SQL (PostgreSQL, BigQuery), TypeScript, Bash.

Machine learning & statistics: neural networks (MLP), Random Forest, fuzzy logic, clustering, statistical downscaling, scenario analysis, entity resolution, Markov decision processes / value iteration, Monte Carlo simulation.

Geospatial & remote sensing: Google Earth Engine, Sentinel-1/2, Landsat, Planet, NAIP, QGIS, ArcGIS, TerrSet, spatial indexing (R-tree), NDVI & change detection, GeoJSON/GeoPackage.

Climate & HPC: CMIP6, GCM/CESM experiments, HPC clusters, Linux/Ubuntu, extreme-event analysis.

Data engineering & cloud: ETL and batch pipelines, Google Cloud Storage, BigQuery, Supabase, REST APIs, Git, AWS, Azure.

TALKS, TEACHING & OUTREACH

- Invited lab talks, ECOPAN Landscape Ecology Lab, Pontificia Universidad Javeriana (2024): "Introduction to GIS Data in R," "Google Earth Engine in Practice," "Finding and Downloading Climate Data."
- Substitute science teacher, Saint George School, Bogotá (Aug. – Nov. 2024): energy, heat, and temperature; evolution and genetics.

LANGUAGES

Spanish — Native · English — C1